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"DEVELOPMENT OF A UNIVERSAL ADJUSTABLE PIPE-CRAWLING INSPECTION AND CLEANING MACHINE"

Thesis submitted in partial fulfillment of curriculum prescribed for the award of the degree of

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IN
MECHANICAL ENGINEERING
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ABSTRACT

Pipelines are widely used in industrial water supply and wastewater management. Over time, the internal surfaces of pipelines may accumulate mud, sediment, scale, and other deposits, resulting in restricted flow and blockages. Conventional inspection and cleaning methods can require manual effort, additional equipment, and difficult access to internal pipe sections. To address these challenges, this project proposes the design and development of a universal adjustable pipe-crawling inspection and cleaning machine for preliminary inspection and controlled cleaning of suitable pipelines.

The proposed machine integrates an adjustable mechanical chassis, a powered wheel-drive mechanism, a camera with LED illumination, a distance sensor, and an audible warning system. The adjustable support mechanism is intended to accommodate different pipe diameters while maintaining stability and wheel contact. The camera provides visual information about the internal pipe condition, while the distance sensor detects nearby obstacles and activates a buzzer when a preset threshold is reached. For cleaning operations, the system incorporates a water-delivery arrangement with a rotating nozzle and a replaceable mechanical cleaning attachment, such as a rotary brush or buffing wheel.

The project involves conceptual design, preliminary engineering calculations, component selection, CAD modelling, fabrication, assembly, and experimental testing. The design considerations include wheel traction, motor torque, adjustment force, electrical power requirements, water pressure, component compatibility, and operational safety. Testing is planned to evaluate mobility, diameter adjustment, inspection visibility, obstacle detection, cleaning effectiveness, and retrieval capability.

The proposed machine aims to provide a compact and adaptable solution for preliminary pipeline inspection and limited deposit removal, reducing the need for direct manual access to suitable pipe interiors. Its practical performance and operating range will be established through controlled prototype testing.

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LIST OF NOTATIONS

Notation	Description
D	Pipe internal diameter (mm)
Dmin	Minimum target pipe internal diameter (mm)
Dmax	Maximum target pipe internal diameter (mm)
r	Wheel radius (m)
n	Wheel rotational speed (rpm)
v	Ideal crawler travel speed (m/s)
F	Force (N)
Frequired	Estimated force required to move the crawler (N)
m	Total crawler mass (kg)
g	Acceleration due to gravity (m/s^2)
Crr	Rolling-resistance coefficient
T	Torque (N·m)
η	Drivetrain efficiency
rp	Pinion pitch radius (m)
Δp	Pressure rise (Pa)
Q	Volumetric flow rate (m^3/s)
P	Power (W)

1. INTRODUCTION

1.1. BACKGROUND

Pipelines are widely used to transport water, wastewater, and process fluids. Over time, internal surfaces can develop corrosion, scale, sediment, grease, mud, and other deposits. Blockages can restrict flow and may cause service interruption. A preliminary visual inspection can help maintenance teams identify the location and apparent condition of a problem before choosing a repair method.

Traditional inspection may require opening access points, inserting a camera manually, or arranging confined-space entry. A compact remotely operated crawler can reduce the need for direct entry during preliminary inspection, provided that the pipe geometry, operating environment, and retrieval arrangements are suitable.

1.2. PROJECT OVERVIEW

The project proposes a universal-style adjustable crawler for selected pipes with an intended internal diameter range of 250–500 mm. The chassis is intended to centre the inspection head and maintain wheel contact as the support arms are adjusted. A camera and light provide visual feedback. A distance sensor is intended to detect nearby obstacles or changes in clearance. Cleaning is proposed through a water jet and a mechanical rotary/buffing tool.

1.3. NEED FOR THE PROJECT

- To support preliminary visual inspection of pipe interiors without routine human entry.
- To provide a compact platform that can be adjusted for more than one pipe diameter.
- To combine inspection and limited cleaning functions in one prototype.
- To create a mechanical design, fabrication, and basic mechatronics.
- To generate repeatable test observations that can guide future improvements.

1.4. SCOPE

The present scope is a prototype for controlled laboratory or demonstration-pipe testing. It is not a certified industrial inspection system and is not intended for live pressurised, hazardous, or gas-carrying pipelines. The stated 150–500 mm range is a design target that must be demonstrated by measurement and testing. Full structural integrity assessment, crack sizing, leak-tight repair, and autonomous navigation are outside the initial scope.

2. LITERATURE AND EXISTING METHODS

2.1. MANUAL INSPECTION

Manual inspection can be effective where the pipe is accessible, but it may require opening the line or entering a confined space. Access, ventilation, contamination, and retrieval are important considerations.

2.2. PUSH-ROD AND CCTV INSPECTION

Push-rod cameras and CCTV crawlers provide visual evidence of internal conditions. Push-rod systems are mechanically simple but can be difficult to advance through long runs or bends. Wheeled crawlers provide powered movement but need a chassis matched to pipe size and surface condition.

2.3. WATER JETTING

Water jetting can loosen deposits and move debris when the pressure, flow, nozzle geometry, and waste collection are suitable. High-pressure water can cause serious injury and can damage pipe liners, joints, or fragile deposits. Pressure rating and safe operating procedures are essential.

2.4. MECHANICAL CLEANING

Rotary brushes, scrapers, and buffing tools can remove selected deposits. Tool selection depends on pipe material, deposit hardness, adhesion, and the risk of damaging the pipe wall. A rigid impact or drilling tool can become stuck or damage the pipe and should not be used until the blockage and pipe condition are understood.

2.5. DESIGN GAP ADDRESSED

The proposed prototype combines an adjustable mechanical support, powered locomotion, camera-based observation, proximity warning, and limited cleaning attachments. The novelty claim should remain modest until a documented prior-art search is completed. A combination of known components alone does not establish patentability.

2.6. COMPARISON

Table 2.1: Comparison of Methods.

Method	Strengths	Limitations	Role in this project
Manual visual inspection	Simple and direct	Access and safety constraints	Reference method only
Push-rod camera	Low mechanical complexity	Limited propulsion and bend capability	Inspection comparison
Powered CCTV crawler	Controlled travel and live view	Cost, diameter-specific chassis	Main design inspiration
Water jet cleaning	Can loosen soft deposits	Pressure hazard; water and waste handling	Controlled cleaning subsystem
Rotary brush/buffing	Useful for selected adhered deposits	Can damage liners or jam	Replaceable low-risk cleaning tool

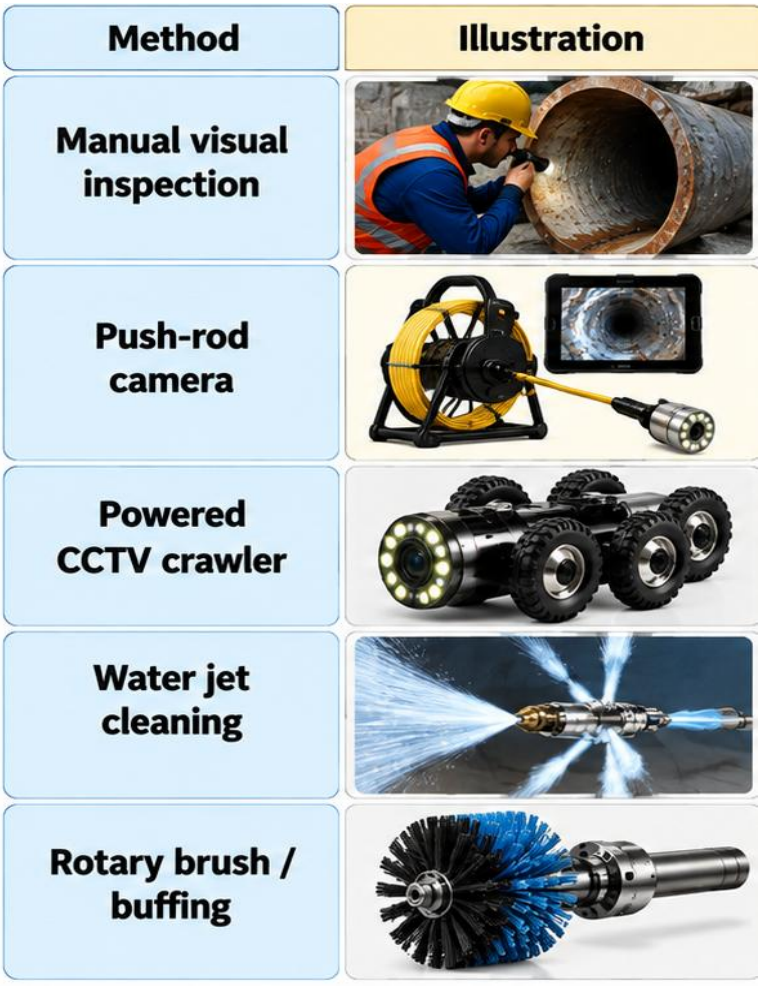


Fig. 2.1: Comparison of Methods.

3. PROBLEM DEFINITION AND OBJECTIVES

3.1. PROBLEM STATEMENT

A practical inspection device should travel inside a pipe, maintain stable contact across a selected range of diameters, provide a useful camera view, identify nearby obstacles, and allow the operator to perform limited cleaning trials. The main engineering challenge is integrating adjustable support, locomotion, inspection, water delivery, and cleaning tools within the available size, power, cost, and safety constraints.

3.2. AIM

"To design and fabricate a wired operated adjustable pipe-crawling device for preliminary internal inspection and controlled cleaning of suitable pipes."

3.3. OBJECTIVES

- Develop an adjustable chassis concept for a target internal diameter range of 150–500 mm.
- Provide driven wheel contact using six locomotion motors, subject to final wheel/leg layout and torque verification.
- Integrate a camera, illumination, distance sensor, and audible warning.
- Provide a water delivery system and a replaceable mechanical cleaning attachment.
- Design mechanical adjustment and locking so the crawler remains stable during movement and tool operation.
- Prepare a fabrication sequence, component list, and realistic cost estimate.
- Develop a repeatable test plan for mobility, inspection quality, obstacle response, cleaning performance, and retrieval.

3.4. DESIGN CONSTRAINTS

The design is constrained by the target budget, pipe diameter and bend geometry, motor torque, available workshop processes, water ingress, cable management, camera visibility, actuator load, and the pressure ratings of the pump circuit. Each constraint must be checked against the final selected parts.

3.5. FUNCTIONAL REQUIREMENTS

Table 3.1: Functional Requirements.

ID	Requirement	Verification method
FR-01	Fit a measured pipe diameter within the selected design range.	Gauge or measure the chassis at minimum and maximum settings.
FR-02	Move forward and backward under operator control.	Travel test in a straight test pipe.
FR-03	Provide live camera observation with illumination.	Check image feed and field of view.
FR-04	Detect a nearby object/clearance change with the distance sensor.	Test at several known distances and surface types.
FR-05	Activate a buzzer when the programmed threshold is crossed.	Record detection distance and warning response.
FR-06	Deliver water to the selected nozzle without leakage.	Low-pressure leak test followed by controlled pressure test.
FR-07	Allow the cleaning tool to be stopped independently.	Functional and emergency-stop test.
FR-08	Remain retrievable if power or drive is lost.	Test the tether/recovery method before entering a long pipe.

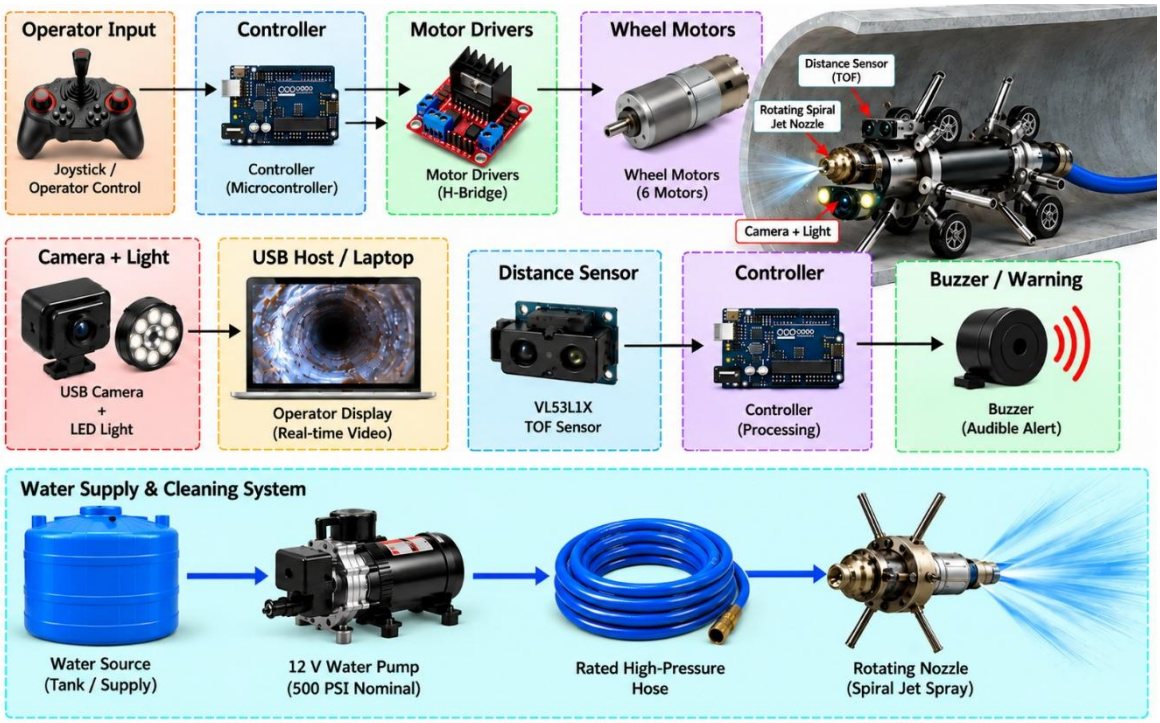


Fig. 3.1: Functional Block Diagram of The Proposed System (Conceptual).

4. CONCEPTUAL DESIGN AND SYSTEM ARCHITECTURE

4.1. OVERALL ARCHITECTURE

The device is divided into five subsystems: (1) adjustable mechanical handicap chassis; (2) locomotion drive; (3) inspection and sensing; (4) cleaning and water delivery; and (5) power, control, and communication. The modules should be arranged so the camera can see ahead, the cleaning tool can be isolated during inspection, and cables and hoses are protected from wheels and rotating parts.



Fig. 4.1: Proposed System (conceptual).

4.2. ADJUSTABLE CHASSIS

The chassis consists of a central body with radial support arms and wheel brackets to maintain stability and contact with the pipe wall. The reference image illustrates the concept only and does not define dimensions or load capacity. The wheel configuration must be finalized, as the proposed six legs with two wheels each conflict with the provisional layout of six driven wheels and motors.

4.3. MECHANICAL ADJUSTMENT

A mechanically actuated rack-and-pinion system is proposed to control arm movement and accommodate different pipe diameters. The design requires guided racks, pinions, bearings, a reduction drive, end stops, and a locking mechanism. Actuation force must be calculated before fabrication.

4.4. LOCOMOTION

Six geared DC motors are provisionally selected. Motor torque and wheel traction must be verified based on crawler mass, wheel radius, rolling resistance, cable and hose drag, pipe inclination, and obstacle loads.

4.5. INSPECTION SUBSYSTEM

A USB camera with LED illumination provides visual inspection. The camera requires suitable splash and impact protection. A microphone may record operational sounds, but audio alone cannot reliably identify blockages without validation.

4.6. OBSTACLE DETECTION

A VL53L1X time-of-flight sensor is proposed for proximity measurement. Its performance must be tested on wet, dark, reflective, and irregular surfaces. A buzzer will provide alerts at a preset distance, supported by additional safety measures.

4.7. CLEANING SUBSYSTEM

The cleaning system combines a water nozzle with a replaceable rotary brush or buffing attachment. A nominal 500-psi pump is proposed, but operating pressure, flow rate, and component ratings must be verified before testing. High-pressure jets must not be used on live pipes or near people without appropriate safety controls.

4.8. MATERIALS AND CORROSION PROTECTION

Steel or aluminium may be used for the frame depending on strength, weight, cost, and fabrication requirements. Shafts, bearings, fasteners, connectors, and seals must be corrosion-resistant and suitable for wet, dirty environments. Protective coatings alone cannot ensure waterproofing.

5. PRELIMINARY DESIGN AND ENGINEERING CALCULATIONS

5.1. PRELIMINARY ASSUMPTIONS

Fig. 5.1: Functional Block Diagram of The Proposed System (Conceptual).

Parameter	Preliminary value/status	Comment
Pipe internal diameter	150–500 mm target	Must be demonstrated with a test gauge.
Supply	12 V DC proposed	A single 12 V, 10 A supply is likely inadequate for every load.
Pump pressure	500 psi nominal	Verify pump curve, flow, hose, nozzle, seals, and fittings.
Drive motors	6 units listed	Motor type, RPM, gearbox ratio, and rated torque not yet specified.
Wheel count	Unresolved	Six wheels in BOM versus twelve wheels in six paired legs in concept notes.
Adjustment stroke	To be designed	Must cover arm geometry and clear the pipe joints.
Crawler mass	Not measured	Required for traction and torque calculations.

5.2. PIPE DIAMETER AND ADJUSTMENT RANGE

For the target diameter range, the mechanism must accommodate a radical change from $D_{min}/2$ to $D_{max}/2$. With D_{min} = 150 mm and D_{max} = 500 mm, the radius changes from 75 mm to 250 mm, giving a radial difference of 175 mm. This is only a geometric envelope; actual actuator stroke depends on arm length, pivot location, and linkage geometry.

$$\text{Radial difference} = (500 - 150) / 2 = 175 \text{ mm}$$

5.3. WHEEL SPEED

For a wheel of radius r rotating at n revolutions per minute, the ideal travel speed is:

$$v = 2\pi rn / 60 \qquad (5.1)$$

Actual travel speed will be lower because of wheel slip, gearbox losses, uneven surfaces, and cable/hose drag. Choose a low controllable speed for inspection and a separate cleaning speed only if safe.

5.4. TRACTION FORCE AND TORQUE

A simplified first estimate of the force needed to move the crawler is:

$$F_{required} \approx C_{rr}\, mg + F_{hose} + F_{cable} + F_{grade} + F_{obstacle} \quad (5.2)$$

where C_{rr} is an assumed rolling-resistance coefficient, m is total mass, g is gravitational acceleration, and the remaining terms represent hose drag, cable drag, slope, and obstacle resistance. Required wheel torque is approximately

$$T = F_{wheel} \times r / \eta, \quad (5.3)$$

where η represents drivetrain efficiency. The load distribution between driven wheels must be considered. Insert measured mass, wheel radius, grade, and drag values before final motor selection.

5.5. MECHANICAL ADJUSTMENT FORCE

The adjustment mechanism must overcome the radial reaction at the wheels, friction in pivots/guides, and any wedging caused by pipe ovality or deposits. A rack-and-pinion system requires pinion torque related to rack force and pinion pitch radius. For a rack force F and pinion pitch radius rp, ideal pinion torque is

$$T = F \times rp; \quad (5.4)$$

gearbox and friction losses increase the required input torque. The arm geometry must be analyzed before selecting a motor or actuator.

5.6. PUMP POWER AND PRESSURE

Hydraulic power can be estimated as

$$P = \Delta p \times Q, \quad (5.5)$$

where Δp is pressure rise in pascals and Q is volumetric flow in cubic meters per second. At 500 psi, pressure is approximately 3.45 MPa. Pump electrical input power must exceed hydraulic output power because of efficiency losses. The flow rate is not specified, so pump power cannot be calculated reliably from pressure alone.

5.7. ELECTRICAL POWER BUDGET

A 12 V, 10 A supply provides a nominal maximum of 120 W before derating. The combined current of six motors, adjustment drives, pump, lights, camera interface, controller, and buzzer may exceed this rating, especially at motor start-up or stall. Use separate, correctly rated power branches, fuses, motor drivers, and a verified supply/battery arrangement. Do not operate all

loads from the listed supply until current draw is measured.

5.8. DESIGN VERIFICATION REQUIRED BEFORE FABRICATION

- Resolve six versus twelve wheels and confirm which wheels are powered.
- Measure total mass and calculate drive torque with a safety factor.
- Confirm whether adjustment uses rack-and-pinion or purchased linear actuators.
- Check actuator force, stroke, speed, duty cycle, end limits, and holding behavior.
- Confirm the pump flow rate and power; match all wetted components to rated pressure.
- Check cable/hose routing through the full adjustment and travel envelope.
- Test turning, recovery, and operation through representative pipe joints and bends.

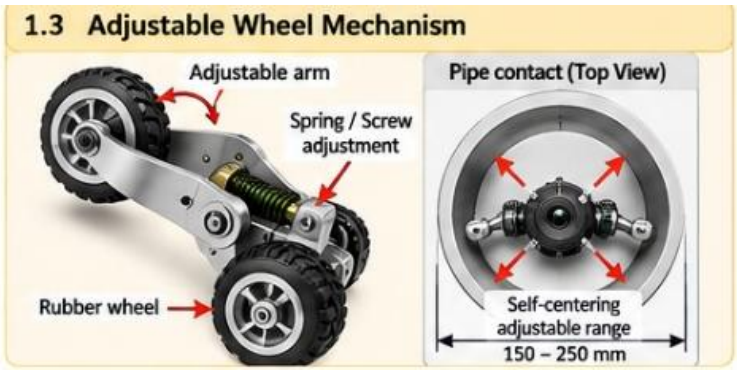




Fig. 5.1: Preliminary Design

6. COMPONENTS AND BILL OF MATERIALS

6.1. ESTIMATED BILL OF MATERIALS

The following costs are based on the user's earlier preliminary list. They are budget estimates only and should be replaced by actual supplier quotations. The rack-and-pinion/adjustment allowance of ₹4,500 is provisional; it is not a verified price for six complete, independent mechanisms.

Table 6.1: Estimated Bill of Materials

No.	Item	Qty.	Cost (₹)	Remarks
1	USB camera/webcam	1	500	Live visual inspection
2	12 V water pump, nominal 500 psi	1	2,000	Water delivery; verify flow and ratings
3	12 V, 10 A SMPS/adapter	1	700	Listed supply; likely needs redesign
4	Control/signal wire (specification to confirm)	Lot	300	Use appropriate conductor and insulation
5	AC cable	Lot	1,000	Keep mains equipment outside wet zone
6	Hose pipe	Lot	800	Pressure rating must exceed operating pressure
7	VL53L1X ToF sensor	1	500	Proximity measurement
8	Flashlight/LED illumination	1	500	Camera lighting
9	Steel pipe/tube for frame	Lot	500	Frame and support members
10	M10 bolts and nuts	Lot	100	Mechanical fastening; verify grade
11	AC switch	1	50	Mains-side control only in protected enclosure
12	Joystick/control switch	1	100	Operator control; final electronics needed
13	Adjustment mechanism	1 set	4,500	Provisional rack-and-pinion/actuation budget
14	Wheels	6	690	Resolve final wheel count and material

15	Metal sheet	Lot	500	Chassis plates and guards
16	Rotating nozzle	1	500	Verify pressure/flow compatibility
17	Buffing/cleaning wheel	1	500	Use suitable brush/tool and guard
18	Geared locomotion motors	6	3,000	Verify rated torque, current, RPM
19	Overhead / contingency	1	3,560	Fabrication, small parts, unexpected costs
Estimated total =			**₹20,300**	

Calculation: listed non-adjustment components total ₹12,240; adjustment allowance ₹4,500; overhead/contingency ₹3,560; total ₹20,300. The amount is not a supplier quotation. It excludes any additional costs arising from a larger power supply, motor drivers, controller, fuses, waterproof connectors, seals, bearings, fabrication labor, test pipe, or a more capable actuator arrangement.

6.2. ITEMS THAT MAY BE MISSING FROM THE BOM

- Controller or microcontroller and suitable USB host/laptop interface.
- Motor drivers sized for motor stall current and control method.
- Power distribution block, fuses, emergency stop, reverse-polarity protection, and suitable connectors.
- Waterproof enclosure, cable glands, seals, strain relief, and corrosion-resistant bearings.
- Recovery tether, retrieval eye, and a method to prevent the crawler being lost in the pipe.
- Pressure-rated hose, fittings, valve, pressure gauge, and safe depressurization arrangement.
- Protective guard and independent switch for the rotating cleaning attachment.

6.3. COST OPTIMISATION

The most effective cost-control step is to simplify the adjustment linkage. A mechanically linked set of arms may reduce the number of actuators, but the linkage must be designed so the crawler remains centered and cannot collapse under load. Do not assume that one actuator can safely drive all arms until its force, stroke, and linkage geometry are calculated.

7. FABRICATION AND ASSEMBLY METHODOLOGY

7.1. DESIGN FINALISATION

Prepare a dimensioned CAD assembly showing the central body, wheel arms, pivots, adjustment drive, motor brackets, camera, sensor, pump line, cleaning tool, guards, and cable/hose paths. Check the minimum and maximum envelope in the target pipe sizes. Perform interference checks at full extension and retraction.

7.2. MATERIAL PREPARATION

Cut frame members and plates to the drawing. Remove sharp edges and verify dimensions before welding or drilling. Use corrosion-resistant fasteners where possible. Avoid placing the camera or electrical enclosure where water jets or debris can strike them directly.

7.3. FRAME FABRICATION

1. Mark and cut the central frame and arm members from the approved drawing.
2. Drill pivot, bearing, motor, and guard mounting holes using jigs where practical.
3. Weld or bolt the frame and check symmetry, squareness, and alignment.
4. Fit wheel brackets and confirm that each wheel can rotate freely.
5. Install the selected adjustment mechanism with mechanical end stops and a locking or holding method.
6. Fit guards around pinions, rotating shafts, and cleaning components.

7.4. MECHANICAL ASSEMBLY

Install the geared motors and wheels, then verify rotation direction and clearance. Route the hose and electrical cable so neither can be caught by the wheels or linkage. Mount the camera and light ahead of the cleaning tool where possible. Install the distance sensor so the intended sensing direction is not blocked by the frame.

7.5. ELECTRICAL ASSEMBLY

Separate low-voltage control wiring from pump/motor power wiring. Fit fuses close to the power source, use suitable wire gauge for measured current, secure all connectors, and provide strain relief. Keep AC mains outside the wet test area. Use a suitable residual-current protective device for mains-fed test equipment.

7.6. WATER CIRCUIT ASSEMBLY

Select a pump, hose, valve, fittings, and nozzle with compatible pressure ratings. Secure the hose against movement and provide a shut-off valve and pressure-relief/depressurization method. First test for leakage at low pressure with the nozzle pointed into a guarded test area. Increase pressure only within the ratings of every component.

7.7. PROPOSED WORKFLOW

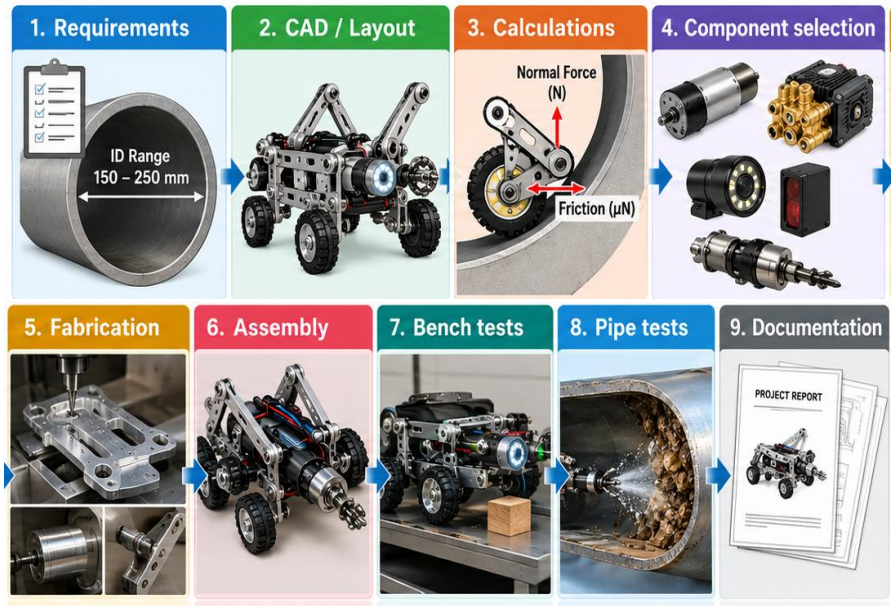


Fig. 7.1: Proposed fabrication and assembly workflow.

7.8. QUALITY CHECKS

- Measure frame dimensions and support-arm symmetry.
- Check fastener tightness, wheel alignment, and actuator end stops.
- Confirm that no cable or hose is pinched through the full adjustment range.
- Check insulation, polarity, fuses, and emergency-stop function.
- Perform a leak test before installing or energizing electronics near the water circuit.

8. CONTROL SYSTEM AND OPERATING PROCEDURE

8.1. CONTROL CONCEPT

The operator controls crawler movement using a joystick or equivalent input through a controller and motor drivers. A USB camera displays live video on a compatible host, while a distance sensor triggers a buzzer when a preset threshold is crossed. The pump and cleaning tool have independent controls.

8.2. PROPOSED CONTROL SEQUENCE

1) Confirm the pipe is isolated, drained, and safe.
2) Inspect the crawler, tether, camera, sensor, wheels, and hose.
3) Adjust and secure the chassis to fit the pipe.
4) Verify controls, video, sensors, buzzer, and motor direction.
5) Insert the crawler slowly with the tether accessible.
6) Inspect obstructions before cleaning.
7) Use approved, low-risk cleaning settings.
8) Stop the pump and tool before retrieval or servicing.
9) Depressurize the hose and isolate power before disconnection.

8.3. SENSOR AND BUZZER LOGIC

If the distance falls below a preset threshold for consecutive readings, the controller slows or stops the crawler and activates the buzzer. Test thresholds and account for invalid sensor readings.

8.4. POWER ARCHITECTURE

Verify the 12 V, 10 A supply against measured operating and peak currents. Use rated motor drivers, fused branches, and a suitable power source. Connect the USB camera to USB port.

8.5. EMERGENCY STOP AND RECOVERY

Provide a physical emergency stop and independent controls for the drive, pump, and cleaning tool. Keep a recovery tether accessible for power or drive failure.

9. SAFETY, RISK ASSESSMENT AND MAINTENANCE

9.1. HIGH-PRESSURE WATER SAFETY

A nominal 500 psi water jet is approximately 34.5 bar and can penetrate skin or cause severe injury. Never point the nozzle toward a person, and never check a jet with a hand. Confirm that the pump, hose, couplings, valves, gauge, and nozzle are rated for the maximum pressure, including transient pressure. Provide a shut-off and depressurization procedure. Use a guarded test area and eye/face protection.

9.2. ELECTRICAL SAFETY

Water and electrical power create a serious shock hazard. Keep mains-powered equipment outside the wet zone. Use properly rated low-voltage equipment, sealed enclosures, cable glands, fuses, strain relief, and suitable residual-current protection on mains-fed test equipment. Inspect cables before every test. Isolate power before cleaning or servicing.

9.3. MECHANICAL SAFETY

Pinch points exist at adjustable arms, rack-and-pinion drives, wheels, and linkages. Rotating buffing tools can catch loose clothing, cable, or debris. Fit guards, use a dead-man or independent stop control where appropriate, and never adjust arms while the mechanism is powered.

9.4. MAINTENANCE

- Rinse and dry the frame after wet testing; do not spray water into bearings or electronics.
- Inspect wheels, tires, bearings, fasteners, and arm pivots for wear or looseness.
- Check actuator alignment, rack/pinion engagement, end stops, and holding function.
- Inspect the hose and fittings for abrasion, swelling, leakage, and pressure-rating labels.
- Clean the camera lens, light cover, and distance-sensor window.
- Record motor current and temperature trends to detect drag or developing faults.
- Replace damaged insulation, seals, or guards before the next test.

9.5. PRELIMINARY RISK ASSESSMENT

Table 9.1: Preliminary Risk Assessment

Hazard	Potential consequence	Control measure
High-pressure jet	Injection injury, eye injury, pipe damage	Pressure-rated components, guarded test, shut-off and depressurization.
Water ingress	Short circuit, shock, equipment failure	Sealed enclosure, low-voltage design, inspection, isolation.
Rotating tool	Entanglement, cuts, tool jam	Guard, independent enable, low-speed trials, isolate before service.
Crawler stuck	Loss of equipment or cable damage	Recovery tether, short test pipes first, avoid untested bends.
Actuator/pinion pinch	Finger injury or mechanism collapse	Guards, end stops, locking, power isolation.
Motor stall	Overheating, gear damage, high current	Current protection, torque margin, stall test, duty-cycle limits.
Sensor error	Missed obstacle or false alarm	Validate on multiple surfaces; operator supervision.
Corrosion/mud	Failure or loss of insulation	Cleaning, drying, corrosion protection, post-test inspection.

10. EXPECTED RESULTS, LIMITATIONS AND FUTURE WORK

10.1. EXPECTED OUTCOMES

The intended outcome is a prototype that can be adjusted to fit selected test-pipe diameters, move under operator control, provide illuminated camera images, give a proximity warning, and demonstrate limited cleaning on a safe test deposit. These are design objectives, not verified achievements until the validation plan is completed.

10.2. LIMITATIONS

- The 150–500 mm diameter range has not yet been verified and may require a substantial chassis and arm stroke.
- The wheel count and leg configuration are not fully resolved in the current component list.
- The initial 12 V, 10 A supply may not support all loads simultaneously.
- Camera visibility may be poor in opaque water, thick mud, or suspended debris.
- The ToF sensor may be affected by wet, dark, reflective, or irregular surfaces.
- High-pressure water cleaning may be unsuitable for fragile pipe materials, joints, or liners.
- Sharp bends, offsets, protruding joints, and heavy solid blockages may trap the crawler.
- The current budget excludes some necessary controls, protection, sealing, and test infrastructure.

10.3. FUTURE IMPROVEMENTS

- Complete CAD kinematic analysis and finite element analysis of the support arms and chassis.
- Use an encoder or motor-current sensing for better travel and stall detection.
- Add a pressure gauge, flow measurement, and a controllable valve to quantify water delivery.
- Develop a quick-change tool head for camera-only inspection, brush cleaning, and nozzle cleaning.
- Improve the camera housing and add recorded distance and test metadata.
- Evaluate a more reliable mechanical centering linkage and a positive locking mechanism.
- Conduct a documented prior-art search before making novelty or patentability claims.
- Test on representative pipe materials, joints, bends, and deposits before any field trial.

11. GANTT CHART

Table 11.1: Gantt Chart – Pipeline Inspection and Cleaning Robot

Project timeline: 10 October 2026 – 15 March 2027

ID	Task / Activity	Oct 2026	Nov 2026	Dec 2026	Jan 2027	Feb 2027	Mar 1–15 2027
1	Finalise requirements & literature review	■					
2	Concept development & selection	■					
3	Preliminary design & CAD modelling	■					
4	Engineering calculations (drive, water jet, structure)		■				
5	Detailed design & component selection (BOM)		■	■			
6	Procurement of components (two-month period)			■	■		
7	Fabrication of chassis and mechanical parts					■	
8	Assembly (mechanical, electrical & plumbing)					■	
9	Software / control programming					■	
10	Individual subsystem testing					■	
11	Integrated testing (land tests)					■	■
12	Pipe environment testing & performance evaluation						■
13	Result analysis & improvements						■
14	Final report, documentation & presentation						■

Key milestones: 12 Oct 2026 – Requirements review complete | 16 Oct 2026 – Concept selected | 27 Oct 2026 – Preliminary design complete | 10 Dec 2026 – Detailed design/BOM complete | 31 Jan 2027 – Procurement period complete | 21 Feb 2027 – Assembly target | 3 Mar 2027 – Integrated testing complete | 8 Mar 2027 – Pipe testing complete | 15 Mar 2027 – Final report and presentation

12. CONCLUSION

The project focuses on the design and development of an adjustable pipeline inspection and cleaning machine intended for use in industrial water and wastewater pipelines. The proposed system combines a pipe-crawling mechanism, adjustable wheel supports, a camera with LED illumination, a time-of-flight distance sensor, an audible warning system, and a cleaning mechanism using a water jet and mechanical cleaning attachment. The main objective is to improve access to the internal sections of pipelines and support the identification and removal of deposits and blockages while reducing the need for direct manual inspection.

The adjustable chassis is intended to accommodate the selected pipe diameter range while maintaining stability and traction during movement. The camera and lighting system provide visual feedback to the operator, while the distance sensor is designed to detect nearby obstacles and activate a warning. The proposed water-delivery and mechanical cleaning systems provide a combined approach for treating suitable deposits. Preliminary engineering calculations for crawler speed, drive torque, adjustment force, water pressure, and electrical power are necessary to guide component selection and verify the feasibility of the design.

The project also highlights the importance of component compatibility, mechanical reliability, electrical protection, pressure-rated water-delivery components, and safe recovery of the crawler. The final performance will depend on the selected motors, adjustment mechanism, power supply, pump flow rate, nozzle configuration, and the type of deposit encountered. Therefore, the design must be verified through fabrication, assembly, and controlled experimental testing before practical industrial use.

Future work includes completing the detailed CAD model, finalizing component specifications, fabricating the prototype, and conducting tests for mobility, diameter adjustment, camera visibility, obstacle detection, cleaning effectiveness, and operational safety. Further improvements may include data logging, improved sealing, interchangeable cleaning tools, and more advanced defect-detection capabilities.

In conclusion, the proposed pipeline inspection and cleaning machine provides a promising engineering concept for improving pipeline maintenance. and safety improvements.

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